

Michigan Electrical Grid Demand Fluctuations

Assessing Data Center Feasibility

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ABSTRACT

The State of Michigan is offering tax incentives to encourage building data centers, although the state has had historical problems with its energy grid. This study analyzes peak energy demand patterns for the state of Michigan to assess whether the additional data center load would be sustainable during high-demand periods. Using publicly available data from MISO Energy and NOAA, this study uses linear regression models trained on 2022-2023 data and tested on 2024 data to predict peak daily demand periods based on seasonal weather conditions, using the Lansing Airport as a representative weather measurement. Key findings include that medium data centers should create minimal vulnerability for the electrical grid, but ten hyperscale data centers could stress the grid by adding 2.3% to peak daily demand in summer months, when average temperature and maximum humidity levels drive demand over 600,000 megawatt-hours per day. The linear regression models in this paper only resulted in an R^2 of 0.68 for warm temperatures and an R^2 of 0.35 for cold temperatures, suggesting polynomial or time-series methods may improve forecasting.

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KEYWORDS

data center, electric grid demand, climate, linear regression, Michigan

1 Introduction

The State of Michigan currently hosts 59 data centers, and DTE and Consumers Energy have recently announced deals in progress to add 6.4 gigawatts of data center power over the next few years and 4 data centers in the Lansing, Ann Arbor, and Kalamazoo markets.[1] A medium data center can consume 10-50 megawatt-hours of power per day, whereas a hyperscale data center can consume 100-1,000 megawatt-hours per day, which is as much as a mid-sized to large city adds to the grid demand.[1] The residents of Michigan are still fighting newly proposed data centers in Kalamazoo and Saline Township, near Ann Arbor, where the proposed data center could demand 1.4 gigawatt-hours per day.[2] Some of the proposed data centers are smaller scale, e.g., the Lansing data center would draw 24 megawatt-hours per day.[3] Data centers require a steady and uninterrupted power supply, as

opposed to the fluctuations inherent to residential seasonal energy demand, which is causing concerns about the source of this steady electricity output.[1]

This paper analyzes under which weather conditions the Michigan electrical grid hits its peak energy demand. The key question is whether the additional load from the proposed data centers would be sustainable for the peak usage times currently experienced across the state. This study helps determine when Consumers Energy and DTE would need to increase their energy output and, perhaps, invest in infrastructure improvements to protect residential customers from the impacts of newly built data centers.

2 Related Work

2.1 Tools and Methods

Multiple studies acknowledge the need for reliable electrical grid demand forecasting or risk analysis, but differ in their tools and methods. Brekken provides metrics and formulae for the quantitative measurement of grid resilience, including risk analysis frameworks.[4] Gilbert *et al.* analyze synthetic data for low voltage network loads and argue that a fusion of peak and conventional forecasts improves continuous ranked probability scores (CRPS) over simple and advanced benchmark methods by up to 9% with no penalty to overall performance.[5] El-Azab *et al.* analyze the New England Control Area electrical demand and suggest short-term seasonal forecasting (Spring, Summer, Autumn, Winter) is the best approach to accurately determine demand.[6] Their analysis included measurements of climate conditions as well as intensity of daily activities, type of day (work day / weekend, holiday), and fuel prices. After testing multiple machine and deep learning algorithms, they determined that the Adaptive Neural-based Fuzzy Inference System (ANFIS) performed the most accurate seasonal forecasting performance overall, based on mean average percentage error (MAPE) values, except for the winter holiday season. This study suggests several avenues for future research for the current study, including adding holiday data and day lengths to future analysis.

2.2 Data and Results

Multiple studies have examined the electric grid's performance under extreme weather conditions by analyzing outage data in various areas, including Michigan. Zhu *et al.* recently quantified electric grid resilience against extreme weather events, using a

spatiotemporal model on power outage data from three service territories on the US East Coast.[7] They conclude that metropolitan or economically strong areas maintain their electrical grids in better condition, and therefore are less vulnerable to extreme weather events. Rural or poorly funded areas require additional planning to prevent or mitigate weather-induced power outages. Do *et al.* performed a similar study on outages between 2018-2020 across 2447 US counties, including in Michigan.[8] The authors noted that precipitation and anomalous heat co-occurred most often with the power outages lasting 8 hours or longer. Snow levels, tropical cyclones, and wildfires were also significant co-occurrences with power outages in some regions. This study also completed a vulnerability analysis on the counties using the Social Vulnerability Index (SVI).[9] Similarly, Rabago and Dutta presented an analysis of the economic impact of a failing power grid to the Michigan House of Representatives in 2023, although their analysis was sponsored by Local Solar for All, and thus had a specific agenda to promote solar energy solutions to Michiganders.[10] In all, the outage studies highlight the importance of well-maintained infrastructure for grid resilience in the faces of unusual storms or wildfires coinciding with high temperatures. They also suggest that analysis of the social or economic impact of outages is a fruitful future direction.

2.3 Contribution of this study

The proposed study undertakes a linear regression model of electrical demand data in Michigan to predict key periods of stress on the energy grid given specific weather patterns. As with El-Azab *et al.*, this study breaks the data into different periods, although in this case, cold and warm temperatures rather than by calendar season. While Zhu *et al.* and Do *et al.* focus on quantifying power outages and identifying vulnerable populations, the current study aims to calculate peak electrical grid usage to identify when the added burden of multiple data centers' power needs might overload the system. The analysis will consider corresponding weather patterns for peak grid demand, instead of focusing on extreme weather events. Power companies will find this analysis more useful to assist their calculations about when additional energy should be produced. A focus on improving grid stability to accommodate data center needs also mitigates the bias toward solar power in the report by Rabago and Dutta.

3 Workflow (= Proposed Work)

To complete the linear regression modeling, I use publicly available data sets, which I cleaned by removing null values, standardizing date formats, and converting the data to the same time periods. The data is warehoused in a simple folder system on a local computer. I visualized the data with Python matplotlib and seaborn and analyzed it using Python statsmodels and pandas. The main questions for this analysis are when the electrical grid experienced peak demand and what weather patterns were correlated with that demand.

3.1 Tools

This project creates a multiple linear regression model using the Python library statsmodels: <https://www.statsmodels.org/stable/index.html> For data manipulation, I use Pandas <https://pandas.pydata.org/docs/index.html> For number crunching and stats tests, I use NumPy <https://numpy.org/> and SciPy <https://scipy.org/> For plotting, I use Matplotlib <https://matplotlib.org/> and seaborn <https://seaborn.pydata.org/>

3.2 Data Sources and Cleaning Steps

3.2.1 Electrical Demand Data

Electrical demand data was obtained via API from EIA: US Energy Information Administration on November 26, 2025. <https://www.eia.gov/odata/index.php>. The data set provides hourly measurements of electrical demand in megawatt hours aggregated across the entire state of Michigan, 2022-2024, from MISO Energy.

The data was relatively clean, with only 7 out of 1096 days missing some of their 24 hours of demand data; I removed the 7 incomplete days from the dataset. There were no other missing or unusual values in the dataset. I also converted the date column from object type to date type; all other data types were in acceptable formats for analysis. Finally, I aggregated the hourly readings into daily measurements to match the daily data in the NOAA dataset. This created 1089 rows of daily electrical demand measurements in megawatt-hours from 2022-2024.

3.2.2 Climate Data

Climate data in .csv format was downloaded from NOAA climate data: <https://www.ncei.noaa.gov/cdo-web/> on Nov. 26, 2025. The dataset focuses on the Lansing, MI Airport as representative of climate and weather for the state of Michigan. This location is appropriate because the proposed data centers will be in Lansing as well as outside of Kalamazoo and Ann Arbor, both of which are also in Central Michigan. The dataset includes daily measurements of temperature, humidity, barometric pressure, precipitation amounts and types, etc. with separate columns for metadata (120 columns in total). A key for the metadata is here: <https://www1.ncdc.noaa.gov/pub/data/ghcn/daily/readme.txt>

This dataset required more significant cleaning than the electrical demand data. Of the 120 columns in the NOAA dataset, 64 lacked data and were removed. A further 14 columns recording air quality conditions and minute-based gusting data fell outside the scope of this study and so were removed. I converted the date column from object type to date type; all other data types were in acceptable formats for analysis.

I chose three methods to fill missing values (NaNs). First, I used linear interpolation, after the data had been sorted by date, to fill singular gaps in the dew point temperature, sea level pressure, station level pressure, and wet bulb temperature on the logic that previous and following days would have had similar readings to the missing values. There were larger gaps in the wind speed values, up to 9 consecutive days. I chose to forward-fill the last known windspeed because windspeed does not rise and fall at

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predictable levels day by day. Finally, in the binary categories recording the presence of fog or drifting snow, I filled NaNs with 0 to indicate the absence of an observation for those days. After this cleaning, there were no more NaN values (except in the metadata columns) and no 9999, the conventional lack of data notation for this dataset.

3.2.3 Data Aggregation and Preparation

With both datasets cleaned and adjusted to daily measurements, I merged climate and electrical data using inner merge on date. The final data shape is 1089 rows (daily measurements) and 43 columns (attributes and metadata). I split this data into a training set covering 2022-2023 and a testing set covering 2024.

3.3 Data Exploration

For the 2022-2023 training data, I first wanted to see the daily electrical demand patterns alongside the daily fluctuations in temperature. Figure 1 plots the daily electrical demand values in megawatt-hours in blue (top) and the range of daily temperatures (bottom), with the red line indicating the daily high temperature, the blue line indicating the daily low temperature, and the gray shading indicating the daily range. There is a visible increase in electrical demand during the summer months, when the temperature highs top 80 degrees Fahrenheit. There is also a slight increase in electrical demand in the winter months, when the temperatures are below 40 degrees Fahrenheit.

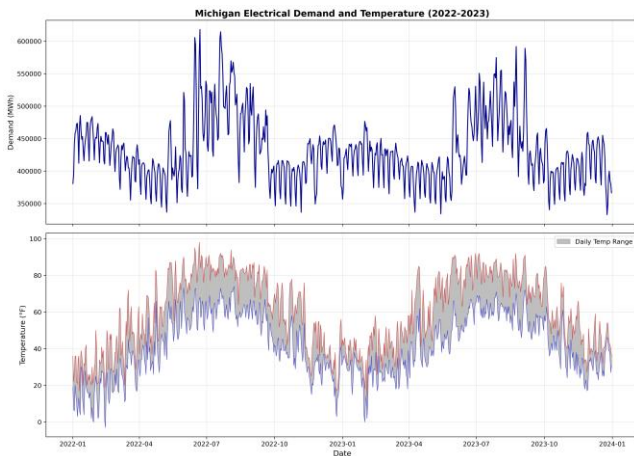


Figure 1: Michigan Electrical Demand (MWh) and Daily Temperature Ranges, 2022-2023.

See Figure 2 for a boxplot visualization of electrical demand by month. The boxplot clearly illustrates the variability in summer electrical demand measurements (June through September) compared to that of the winter months.

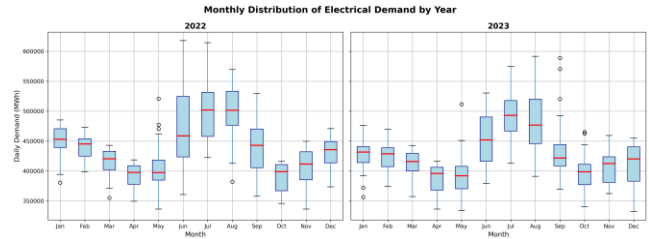


Figure 2: Boxplot of Daily Electrical Demand Measurements (MWh) by Month, 2022-2023.

Because the training dataset had 18 possible weather patterns as predictor (independent) variables for the electrical demand result (dependent) variable, I wanted to check the correlations between the main weather variables. Figure 3 shows the correlation matrix between temperature, humidity, pressure, precipitation, and windspeed variables. The binary observations of fog type and drifting snow are omitted from this matrix.

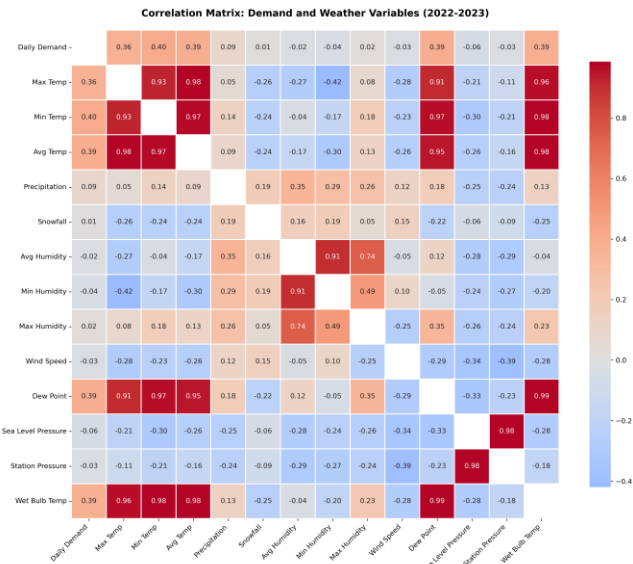


Figure 3: Correlation Matrix between Electricity Demand (MWh) and Weather Variables, 2022-2023.

Unsurprisingly, there is a strong correlation between the average, maximum, and minimum temperatures, but also between temperature and the dew point and wet bulb temperature measurements. Thus, only one of these variables should appear in any regression model. The humidity measurements are similarly correlated, so only one should be chosen for any model. Likewise, only one pressure measurement should be chosen for any regression modeling.

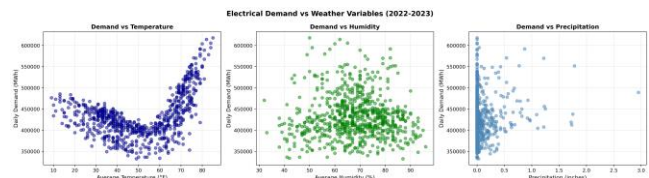


Figure 4: Scatterplots Showing Electricity Demand (MWh) vs. Average Temperature (degrees Fahrenheit), Average Humidity (percent) and Precipitation (inches)

Finally, I plotted the electricity demand against three weather variables: average temperature, average humidity, and precipitation to see if obvious patterns emerged. A significant bifurcation of data points appeared in the electricity demand v. average temperature scatterplot. While the overall data is not linear, if I separate the datasets at 55 degrees Fahrenheit, two linear relationships form. Thus, I will split my linear regression models into cold season data ($< 55^{\circ}\text{F}$) and warm season data ($\geq 55^{\circ}\text{F}$). The distribution of the overall electrical demand data is not clearly normal (Figure 5), but the tails of the warm and cold season data are less lopsided (Figure 6).

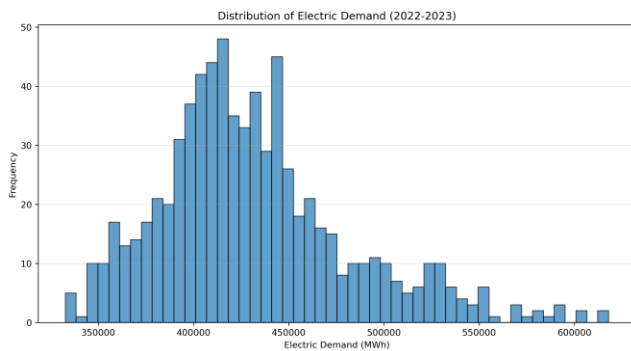


Figure 5: Distribution of Michigan Daily Electrical Demand, 2022-2023

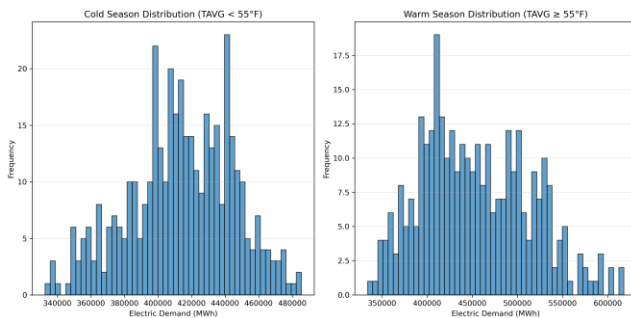


Figure 6: Distribution of Michigan Daily Electrical Demand, split into a Cold Season (Average Temperature $< 55^{\circ}\text{F}$) and a Warm Season (Average Temperature $\geq 55^{\circ}\text{F}$).

3.4 Linear Regression Preparation

To fit linear regression models, the electrical demand data has been split into a cold season (average temperature $< 55^{\circ}\text{F}$) and a warm season (average temperature $\geq 55^{\circ}\text{F}$). The first step is to select the appropriate weather predictors (independent variables) for the electrical demand result (dependent variable) for each season.

3.4.1 Model Selection

Using Python's `itertools` combinations function, I fit all possible models and compared their Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), coefficient of determination (R^2), and Adjusted R^2 values to find the best set of predictors for the two linear regression models (cold season and warm season). This took three iterations.

The first iteration attempted to run with all 18 possible weather variables, but that would require testing $2^{18} - 1 = 262,143$ combinations. It took 20+ minutes to run per season. The precision was not worth the lack of efficiency. To simplify, I removed minimum and maximum humidity measurements, sea level pressure, dew point, and wet bulb temperature, because these measurements are correlated with other measurements in the dataset. The resulting models returned a 4-way tie in AIC values for the potential cold season models and a 10+ way tie in AIC values for the potential warm season models.

On the third iteration, I removed the binary classifiers for fog, rime, and drifting snow, because certain cold weather exclusive phenomena were not appearing in top models for the cold season but they were appearing in top models for the warm season, which indicates correlation complications. I added the variables for minimum and maximum humidity measurements, sea level pressure, dew point, and wet bulb temperature, assuming the combination function would sort out any multicollinearity problems. The single cold season model with the lowest AIC value included the following predictors: average temperature (Fahrenheit), precipitation (inches), minimum humidity (percentage), average wind speed (miles per hour), and average station level pressure (hectopascals*10). (AIC: 9269.98, BIC: 9293.96, R^2 : 0.393, Adj. R^2 : 0.385). According to the correlation matrix in Figure 3, these variables had acceptable correlation levels to keep them all in the model.

For the third iteration, the top four models from the AIC test for the warm days came back with a tie again. The simplest of these models had average temperature, minimum temperature, and average dew point, which are all closely correlated temperature metrics. Thus, I tested which temperature metric, as a sole predictor, resulted in the best model by AIC. Average temperature (AIC: 7771.82, R^2 : 0.639) outperformed minimum temperature or maximum temperature as the only predictor. Next, I added second predictors to average temperature (precipitation, average humidity, average wind speed, average station level pressure.) When average humidity performed best (AIC: 7751.54, R^2 : 0.663), I tested whether maximum humidity or minimum humidity would perform even better. Ultimately, the best model predictors for the warm season are average temperature and maximum humidity (AIC: 7749.58, R^2 : 0.665).

One limitation is that the condition number is high for both models, perhaps because of the natural correlations between weather conditions. It will be future work to refine the models to remove as much multicollinearity as possible without losing predictive power. Another limitation, specifically with the cold

season model, is that the predictor for minimum humidity had a p-value of 0.06, which suggests that it might not be statistically significant. But removing this predictor raised the AIC value from 9270 to 9272 and the adjusted R^2 shrank from 0.385 to 0.381, so the model retained this predictor for now. A future analysis might drop this predictor and perform further measurements to evaluate goodness of fit.

4 Evaluation

This study trained linear regression models on data for two years (2022-2023) and tested it on data from another year 2024. These years were chosen because they are the most recent, complete years of data. The training data was split into a cold season (average temperature $< 55^\circ\text{F}$) and a warm season (average temperature $\geq 55^\circ\text{F}$) to model the two linear relationships observable in the data. Then, I used AIC and R^2 values to choose a viable model for the cold temperatures and the warm temperatures. Evaluation of these two models shows that the cold season model predicted 2024 values with 5% accuracy and a prediction R^2 of 0.35, as compared to the warm season model's 6% accuracy and prediction R^2 of 0.68. For the purposes of the Data Modeling course project, this is adequate, but future work should include comparing the linear models to other methods and/or including additional predictors to better model the demand variable.

4.1 Cold Season Linear Regression Model

The cold season model with the best AIC included the following predictors: average temperature (Fahrenheit), precipitation (in inches), minimum humidity (percentage), average wind speed (miles per hour), and average station level pressure (hectopascals*10). Nonetheless, the AIC and BIC values were high (AIC: 9269.98, BIC: 9293.96) and the R^2 values were not as promising as those for the warm season model (R^2 : 0.393, Adj. R^2 : 0.385).

The linear regression model formula for days with average temperatures below 55 degrees Fahrenheit (what I am calling the cold season) is:

$$\begin{aligned} y_{\text{DemandMWh}} = & -87,270 - 1,681.54 \cdot x_{\text{AvgTemp}} \\ & + 17,420 \cdot x_{\text{Precip}} + 180.22 \cdot x_{\text{MinHumid}} \\ & + 1,217.07 \cdot x_{\text{AvgWind}} + 18,550 \cdot x_{\text{StationLvlPressure}} \end{aligned}$$

The intercept (-87,270) is not practically meaningful because it represents conditions outside of the observed data range, i.e. the weather predictors will never all be zero as a real-world scenario. Otherwise, the coefficients indicate that, holding all other variables constant:

- A 1°F increase in average temperature is associated with a 1,681.54 MWh decrease in peak demand. This corresponds with the pattern that, as temperatures warm to 55 degrees on average, the daily electric demand decreases. The average temperature predictor had a p-value of 0.000, indicating it is statistically significant.

- A 1 inch increase in daily precipitation is associated with a 17,420 MWh increase in peak demand. The precipitation predictor had a p-value of 0.012, indicating it is statistically significant.
- A 1% increase in the minimum humidity is associated with a 180.22 MWh increase in peak demand. The minimum humidity predictor had a p-value of 0.06, indicating it might not be statistically significant, but removing it from the model raised the AIC value and dropped the R^2 of the model.
- A 1 mile per hour increase in the average daily windspeed is associated with a 1,217.07 MWh increase in peak demand. The windspeed predictor had a p-value of 0.001, indicating it is statistically significant.
- A 1 hectopascal (*10) increase in the station level pressure is associated with a 18,550 MWh increase in peak demand. The station level pressure predictor had a p-value of 0.006, indicating it is statistically significant.

Figure 7 visualizes the cold season electricity demand for 2022-2023 with its linear fit demonstrating the downward trend in demand as the temperature increases. The cold season model overall showed an R^2 of 0.393 and adjusted R^2 of 0.385, which indicates that only 38.5% to 39.3% of the electricity demand is explained by the weather variables. Future work needs to include additional data, such as length of daylight hours or calendar day type to explain more of the variations in electricity demand. Furthermore, the condition number for this model is 12,200, which suggests extremely high collinearity between the predictors, which needs to be addressed in future work.

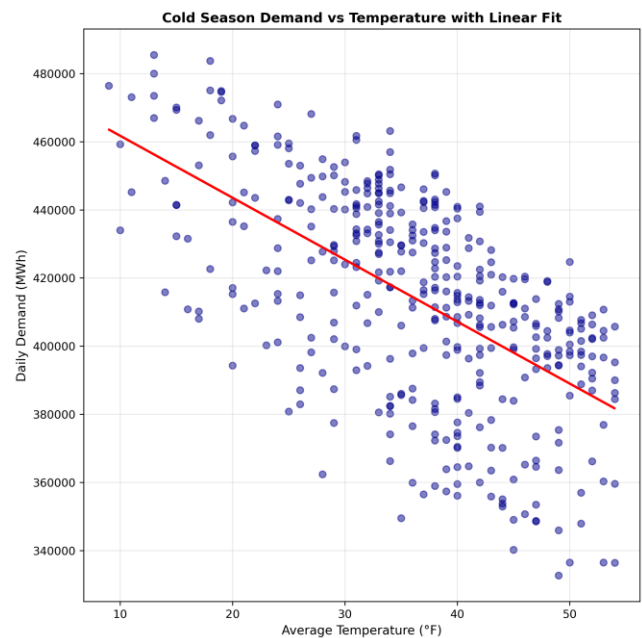


Figure 7: Cold Season Electricity Demand v. Temperature with a linear fit

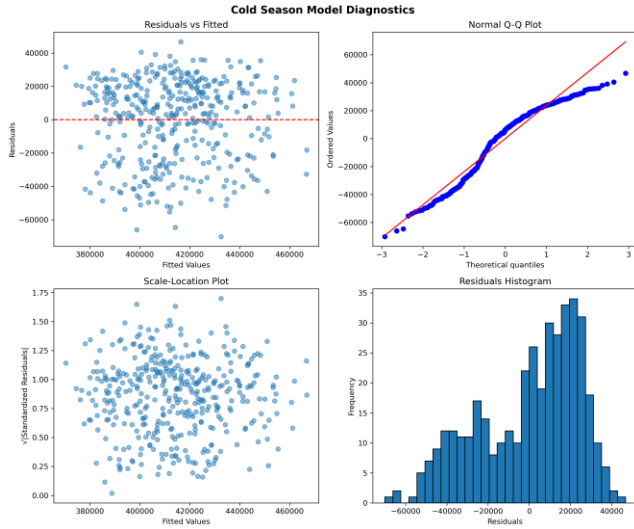


Figure 8: Diagnostic Plots for the Cold Season Linear Regression Model

Diagnostic plots for the cold season model (Figure 8) demonstrate that the model reasonably meets the assumptions of a linear model but might have trouble predicting extreme demand. The residuals v. fitted plot (top left) shows random scatter around the horizontal 0 line, which indicates adequate linearity. There is no discernible funneling, which indicates reasonably constant variance (homoskedasticity). Similarly, the scale-location plot (bottom left) also shows no upward or downward trend in the scatter, which confirms that the homoskedasticity is acceptable. The quantile-quantile plot (top right) shows a slight S curve; the heavy tails mean that there might be extreme outliers, but much of the data is normally distributed. The residuals histogram (bottom right) confirms that the data is generally normally distributed, but with some variations. Therefore, the linear model for the cold season should hold fairly well for predicting typical conditions but might not perform well at the extremes.

When used to predict the demand in MWh for 2024, the cold season model performed poorly (R^2 of 0.35) compared to the warm season model (R^2 of 0.68). See Figure 11 for actual 2024 values v. the prediction line for the two models. The root mean squared error (RMSE) was 24,754.41 MWh and the mean absolute error (MAE) was 20,222.54 MWh. This means that for a demand range of appx. 340,000 to 480,000 MWh, the model's prediction errors were appx. 5% of actual demand values. This is not surprising given the R^2 of 0.35. As mentioned above, additional predictors such as calendar day or length of daylight hours might improve the model.

4.2 Warm Season Linear Regression Model

The best model for the warm season included average temperature and maximum humidity as predictors (AIC: 7749.58, R^2 : 0.665).

The linear regression model formula for days with average temperatures at or above 55 degrees Fahrenheit (what I am calling the warm season) is:

$$y_{DemandMWh} = -126,300 + 7,192.40 \cdot x_{AvgTemp} + 1,033.75 \cdot x_{MaxHumid}$$

The intercept (-126,300) is not meaningful because it represents conditions outside of the observed data range, *i.e.* the impossible scenario that temperature and humidity would reach zero in the warm season. Otherwise, the coefficients indicate that, holding all other variables constant:

- A 1°F increase in average temperature is associated with a 7,192.40 MWh increase in peak demand. The average temperature predictor had a p-value of 0.000, indicating it is statistically significant.
- A 1% increase in maximum humidity is associated with a 1,033.75 MWh increase in peak demand. The maximum humidity predictor had a p-value of 0.000, indicating it is statistically significant.

Figure 9 visualizes the warm season electricity demand for 2022-2023 with its linear fit demonstrating the upward trend in demand as the temperature increases. The warm season model overall showed a R^2 of 0.665 and adjusted R^2 of 0.663, which indicates that 66-67% of the electricity demand is explained by the weather variables. Future work could include additional data, such as calendar day type to explain more of the variations in electricity demand. Furthermore, the condition number for this model is 1,540, which suggests high collinearity between the predictors, which needs to be addressed in future work.

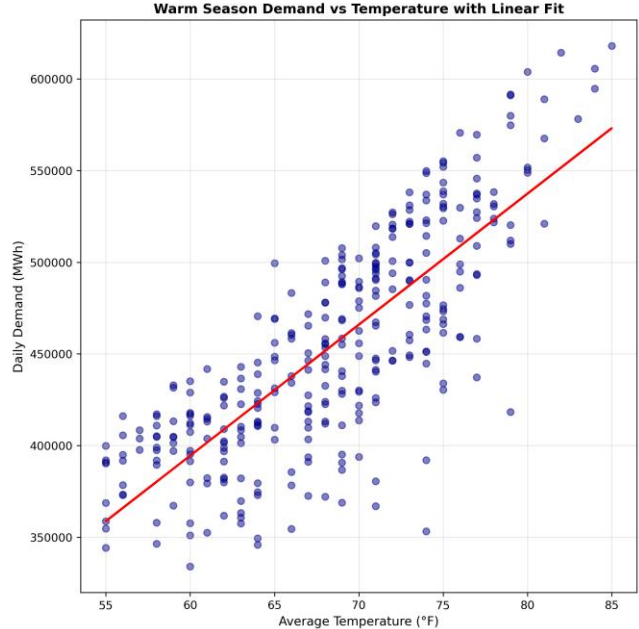


Figure 9: Warm Season Electricity Demand v. Temperature with a linear fit

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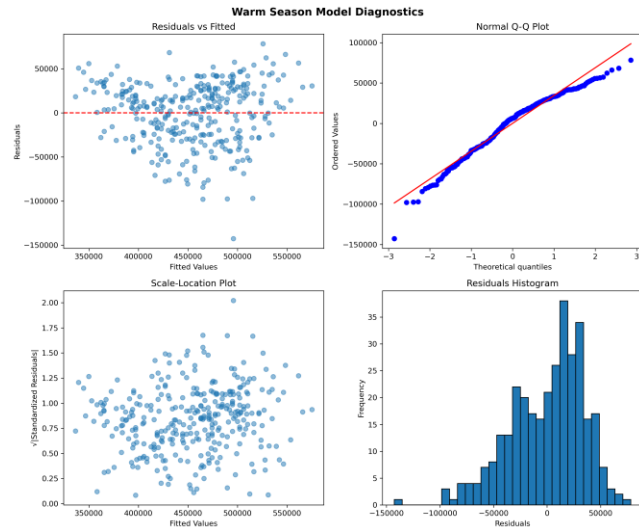


Figure 10: Diagnostic Plots for the Warm Season Linear Regression Model

As with the cold season model, the diagnostic plots for the warm season model (Figure 10) demonstrate no major problems with the assumptions for a linear model, except for some issues predicting extreme demands. The residuals v. fitted plot (top left) shows random scatter, which indicates adequate linearity. There is no major funneling, which indicates reasonable homoskedasticity. This is confirmed by the scale-location plot (bottom left). The quantile-quantile plot (top right) shows a slight curve with the tails deviating from the reference line, which means that predictions for unusually high or low demand days will have larger errors. The residuals histogram (bottom right) confirms that the data is generally normally distributed. Therefore, the linear model for the warm season should predict typical conditions well but might be less reliable when predicting extreme values.

In predicting the demand in MWh for 2024, the warm season model performed reasonably well (R^2 of 0.684) compared to the cold season model (see Figure 11). The root mean squared error (RMSE) was 34,412.48 MWh and the mean absolute error (MAE) was 27,806.15 MWh (6.2% of typical demand for a range of appx. 350,000 to 550,000 MWh). These metrics reflect the larger variability in summer demand for electricity, but also the greater correlation between average temperatures plus humidity and the need for air conditioning, which raises electrical demand.

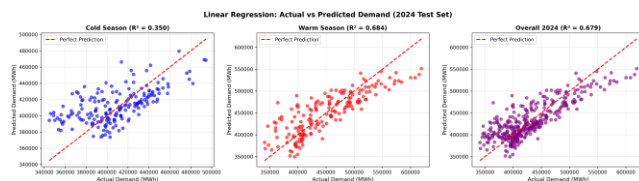


Figure 11: Testing of the Linear Regression Models for the Cold Season, the Warm Season, and Combined on the 2024 Testing Data

When combining the two models' predictions, the overall plot shows points clustering along the diagonal at the lower end, which indicates that predictions are generally unbiased in that range. At the uppermost range (525,000-650,000 MWh), however, the warm season model consistently underpredicts demand, which is counterproductive to the original questions of this study. Future work needs to refine the model to better predict extreme values, perhaps by adding predictors accounting for weekends and holidays. Other improvements might be to model the data using a time-series analysis such as the SARIMAX (seasonal autoregressive integrated moving average with expounding factors) method or to split the data not by average temperature but by actual calendar season.

5 Discussion

5.1 Project Timeline

For the Data Mining course project, the timeline was as follows:

- Oct. 21-Nov. 11, 2025: project ideation and research
- Nov. 18, 2025: design data warehousing folders
- Nov. 18-26, 2025: locate and download data. Formulate strategy to deal with data on different temporal and spatial levels.
- Nov. 26-29, 2025: explore and visualize data. Create a solution for the apparent dual linear relationship between temperature and electrical demand. Consider adding a SARIMAX analysis.
- Nov. 30, 2025: Determine appropriate linear regression predictors for warm and cold temperatures. Train and test linear regression models.
- Dec. 2-9, 2025: Analyze the linear regression models, decide to push SARIMAX or other model refinements to future work sections.

5.2 Project Findings, Challenges, and Alternatives

The key questions of this study were when the Michigan electrical grid experienced peak demand and what weather patterns were correlated with that demand. The goal is to enable energy companies to plan when they would have to increase supply to meet the usual demand plus the constant, additional load required for 4-10 new data centers in Michigan. While the data centers are still under negotiation, we can posit some preliminary metrics of their potential demand. Assuming 4 data centers of medium size, in keeping with the current data center size in Michigan, we could expect a daily additional load of approximately 50 MWh per data center, or 200 MWh additional load per day. Given the current demand range of 350,000 to 600,000 MWh per day, four medium-sized data centers would pose no challenge for the overall electrical grid. On the other hand, if the data centers were as large as the beleaguered Saline Township data center, which was expected to draw 1.4 gigawatt-hours per day, and if there were ten such data centers, then the daily additional demand would reach 14 gigawatt hours, or 14,000 megawatt hours. This represents a 2.3% extra load on the heaviest load days in June and July, when the grid already demands over 600,000 megawatt-hours of

electricity. The next question is whether the models presented in this paper were able to predict which weather conditions would lead to the highest levels of demand.

This study tested the predictive power of a linear regression model that split the parabolic demand data into two, roughly linear sections around the average temperature of 55 degrees Fahrenheit. The colder temperature model used average temperature (Fahrenheit), precipitation (in inches), minimum humidity (percentage), average wind speed (miles per hour), and average station level pressure (hectopascals*10) as predictors, although there is multicollinearity between these variables that needs to be removed in future work on this project. This cold temperature model only predicted the 2024 data with an R^2 of 0.35. The warm temperature model, on the other hand, used average temperature and maximum humidity as predictors, and was able to predict the 2024 demand patterns with an R^2 of 0.68. Unfortunately, the model underpredicted demand at levels over 525,000 megawatt-hours, which is the exact region where the prediction is most needed. I think this is due to the non-normal distribution of the data (see Figure 5), where there was an abnormally large tail in the larger demand levels to the right side of the distribution, exactly where the model is predicting least well.

Both models suffered from significant multicollinearity as well. I attempted to remove some predictors, such as minimum humidity from the cold season, which seemed not to be statistically significant and might have been adding to the multicollinearity. Each removal made the AIC and R^2 values worse, though, so I made the choice to keep the more complex models. I have learned that there are many correlating factors in weather data that might be impossible to completely remove. In future work on this project, I could try more iterative testing to find which models appropriately balance model complexity with predictive accuracy.

Based on these challenges, the linear regression models need significant refinement. Possibilities include adding additional predictors from data on whether a date was a holiday, weekday, or weekend or what the length of daylight hours were for each day. Another approach could be comparing the effectiveness of the linear regression method with a time-series analysis such as SARIMAX, which could capture the cyclical nature of the data, or a polynomial regression method, which would better model the distribution of the data as a whole.

An additional challenge to the current study involved the differing scales of the publicly available data. On the one hand, the NOAA climate data provided daily averages for weather conditions at many individual locations throughout the state of Michigan. I chose Lansing Airport as a representative location for the climate of central Michigan, where most of the new data centers will be built. On the other hand, the EIA electricity data provided hourly demand values for the entire state, including the Upper Peninsula. For this study, I chose to aggregate the demand data into daily readings to match the climate data. It would be instructive in the future to obtain datasets that are hourly across both electrical and climate data and have finer-grained locational data. This data

should result in more precise predictions of the electrical demand on individual substations. If the proposed data centers are built, the results of the model can then be validated against actual data center impact on the energy infrastructure.

6 Conclusion

This study analyzes peak energy demand patterns for Michigan's electrical grid to assess whether adding 4-10 medium to hyperscale data centers would be sustainable during high-demand periods. Using publicly available data from MISO Energy and NOAA (2022-2024), linear regression models were trained on 2022-2023 data and tested on 2024 data to predict peak daily demand based on weather conditions at Lansing Airport.

Key findings include that peak energy demand occurs between June and September, during spikes in average temperature and maximum humidity readings. Medium data centers should create minimal vulnerability for the electrical grid, but ten hyperscale data centers could stress the grid by adding 2.3% to peak daily demand in the summer months, when average temperature and maximum humidity levels drive demand over 600,000 megawatt-hours per day. The linear regression models in this paper only resulted in an R^2 of 0.68 for warm temperatures and an R^2 of 0.35 for cold temperatures, suggesting polynomial or time-series methods or splitting the data by season rather than temperature may improve forecasting.

This study helps Consumers Energy and DTE anticipate when additional generation capacity will be needed, specifically during summer heat waves with high humidity. Refining the model with calendar data or daylight estimates will narrow down the vulnerability window during these months. If the proposed data centers are built, the results of the model can then be validated against actual data center impact on the energy infrastructure.

Another fruitful avenue of analysis would be to account for which Michigan neighborhoods have the oldest infrastructure, using neighborhood age as a proxy for potentially older energy grid components, and which neighborhoods are farthest from a main transformer. Neighborhoods with older infrastructure or at a large distance from a transformer might be at heightened risk of an outage when the electrical grid is overloaded. Geospatial analyses can also be expanded to include vulnerable populations using the Social Vulnerability Index.

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